

GAUTAM KANNA

Brampton, Ontario | 647-864-4259 | gautamkanna.exp@gmail.com | github.com/Gautam-Kanna/crypto-momentum-reversal

QUANT PROJECTS

Crypto Momentum & Reversal Statistical Arbitrage

Sep 2026 - Oct 2026

- Created a **cross-sectional long/short market-neutral strategy** on 10 large-cap coins (BTC, ETH, BNB, XRP, ADA, DOGE, SOL, DOT, LTC, LINK) trading 4-hour bars with realistic **20 bps trading costs (7 bps with limit orders)**
- Researched **time-horizon-dependent patterns** across 2021–2023 training period: identified **mean reversion at 4h** and **momentum strategy from 8h–30d**; enhanced signals with **volume filtering**, **residual returns** (beta-adjusted to Bitcoin), and **machine learning models**; tested **regime detection** and seasonality
- Out-of-sample performance (2024–2025, never seen before): **9.8% annual return, 0.745 Sharpe ratio, -12.3% max drawdown, -0.015 beta to BTC; alpha uncorrelated to market direction**
- Implemented end-to-end in **Python (Pandas, NumPy, Matplotlib)**; built **automated data pipeline from Binance API**; used **exponential smoothing** to reduce portfolio turnover and improve net returns

EDUCATION

Bachelor of Science, Computer Science

York University | Sep 2017 - Aug 2021 | Toronto, Ontario

Dean's List (multiple semesters) | Courses: **Linear Algebra (A)**, **Engineering Statistics (A)**, Applied Multivariate & Vector Calculus (A)

PROFESSIONAL EXPERIENCE

Software Developer, Hamsi Cloud 9 Inc

Jul 2021 - May 2022 | Brampton, Ontario

- Designed and deployed a **MySQL database managing 10,000+ product records and 15,000+ customer profiles**; **reduced data retrieval latency by 30%** through schema optimization and indexing
- Engineered **automated batch-based data synchronization** between backend systems and web platform; **eliminated 80% of manual data-entry errors** via scheduled processing and validation pipelines

QUANT SKILLS

Programming: Python (Pandas, NumPy, Matplotlib), Java, SQL

Specialties: Statistical arbitrage, backtesting, signal research, machine learning, factor analysis, **mean reversion**, **momentum strategy design**, **regime detection**, **risk management**, data pipeline architecture, portfolio optimization, algorithmic trading

Tools: Jupyter, Git, Binance API

INTERESTS

Chess, Trading, Meditation